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Manifold for turbine blade of gas turbine engine

Abstract

A turbine blade includes a platform, a blade airfoil attached to one side of the platform, a blade root attached to the other side of the platform, a manifold attached to the blade root. The blade root defines a first cooling passage and a second cooling passage. The manifold includes an outer wall, a first compartment having a first flow area defined by a first aperture formed at the outer wall, and a second compartment having a second flow area defined by a second aperture formed at the outer wall. The first compartment meters a non-zero first flow of cooling air to the first cooling passage through the first flow area. The second compartment meters a non-zero second flow of cooling air to the second cooling passage through the second flow area that is different than the first flow area.

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Background/Summary

BACKGROUND

(1) A gas turbine engine typically includes a compressor section, a turbine section, and a combustion section disposed therebetween. The compressor section includes multiple stages of rotating compressor blades and stationary compressor vanes. The combustion section typically includes a plurality of combustors. The turbine section includes multiple stages of rotating turbine blades and stationary turbine vanes. Turbine blades and turbine vanes often operate in a high temperature environment and are internally cooled.

BRIEF SUMMARY

(2) In one aspect, a turbine blade includes a platform defining a first side and a second side. The turbine blade also includes a blade airfoil attached to the first side of the platform, the blade airfoil includes a pressure side wall and a suction side wall defining a blade airfoil interior. The turbine

blade also includes a blade root attached to the second side of the platform, the blade root including a blade root leading edge and a blade root trailing edge with respect to a flow direction of a working flow, the blade root defining a first cooling passage and a second cooling passage that are in flow communication with the blade airfoil interior. The turbine blade also includes a manifold attached to the blade root, the manifold being non-planar, the manifold includes an outer wall, a first compartment having a first flow area defined by a first aperture, the first aperture being formed at the outer wall and open to the first compartment, the first compartment metering a non-zero first flow of cooling air to the first cooling passage through the first flow area, and a second compartment having a second flow area defined by a second aperture, the second aperture being formed at the outer wall and open to the second compartment, the second compartment metering a non-zero second flow of cooling air to the second cooling passage through the second flow area that is different than the first flow area.

(3) In one aspect, a manifold for use with a turbine blade, the manifold includes a first side plate, a second side plate, an outer plate that extends between the first side plate and the second side plate, a forward plate attached to the first side plate, the second side plate, and the outer plate at one end, an aft plate attached to the first side plate, the second side plate, and the outer plate at the other side that is opposite to the one end, a first compartment having a first flow area defined by a first aperture, the first aperture being formed at the first side plate and open to the first compartment, a second compartment having a second flow area defined by a second aperture, the second aperture being formed at the first side plate and open to the second compartment, the second flow area being different than the first flow area.

(4) In one aspect, a manifold for use with a turbine blade, the manifold includes a first compartment having a first compartment outer wall, the first compartment outer wall defining a first aperture that is open to the first compartment, the first aperture defining a first flow area, and a second compartment having a second compartment outer wall, the second compartment outer wall defining a second aperture that is open to the second compartment, the second aperture defining a second flow area that is different than the first flow area.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced.
- (2) FIG. 1 is a longitudinal cross-sectional view of a gas turbine engine taken along a plane that contains a longitudinal axis or central axis.
- (3) FIG. 2 is a perspective view of a turbine blade for use with the gas turbine engine shown in FIG. 1.
- (4) FIG. 3 is a perspective view of a manifold for use with the turbine blade shown in FIG. 2.
- (5) FIG. 4 is a perspective exploded view of the manifold shown in FIG. 3.
- (6) FIG. 5 is a perspective view of a manifold suitable for use with the turbine blade shown in FIG. 2.
- (7) FIG. 6 is a perspective view of a manifold suitable for use with the turbine blade shown in FIG. 2.
- (8) FIG. 7 is portion of a perspective view of the turbine blade shown in FIG. 2 that better illustrates the blade root and the manifold shown in FIG. 3.
- (9) FIG. 8 a perspective view of a manifold suitable for use with the turbine blade shown in FIG. 2.
- (10) FIG. 9 a perspective view of a manifold suitable for use with the turbine blade shown in FIG. 2.
- (11) FIG. 10 is portion of a perspective view of the turbine blade shown in FIG. 2 that better

illustrates the blade root and the manifold shown in FIG. 8.

DETAILED DESCRIPTION

(12) Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in this description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

(13) Various technologies that pertain to systems and methods will now be described with reference to the drawings, where like reference numerals represent like elements throughout. The drawings discussed below, and the various embodiments used to describe the principles of the present disclosure in this patent document are by way of illustration only and should not be construed in any way to limit the scope of the disclosure. Those skilled in the art will understand that the principles of the present disclosure may be implemented in any suitably arranged apparatus. It is to be understood that functionality that is described as being carried out by certain system elements may be performed by multiple elements. Similarly, for instance, an element may be configured to perform functionality that is described as being carried out by multiple elements. The numerous innovative teachings of the present application will be described with reference to exemplary non-limiting embodiments.

(14) Also, it should be understood that the words or phrases used herein should be construed broadly, unless expressly limited in some examples. For example, the terms “including”, “having”, and “comprising”, as well as derivatives thereof, mean inclusion without limitation. The singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Further, the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. The term “or” is inclusive, meaning and/or, unless the context clearly indicates otherwise. The phrases “associated with” and “associated therewith” as well as derivatives thereof, may mean to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, or the like. Furthermore, while multiple embodiments or constructions may be described herein, any features, methods, steps, components, etc. described with regard to one embodiment are equally applicable to other embodiments absent a specific statement to the contrary.

(15) Also, although the terms “first”, “second”, “third” and so forth may be used herein to refer to various elements, information, functions, or acts, these elements, information, functions, or acts should not be limited by these terms. Rather these numeral adjectives are used to distinguish different elements, information, functions or acts from each other. For example, a first element, information, function, or act could be termed a second element, information, function, or act, and, similarly, a second element, information, function, or act could be termed a first element, information, function, or act, without departing from the scope of the present disclosure.

(16) Also, in the description, the terms “axial” or “axially” refer to a direction along a longitudinal axis of a gas turbine engine. The terms “radial” or “radially” refer to a direction perpendicular to the longitudinal axis of the gas turbine engine. The terms “downstream” or “aft” refer to a direction along a flow direction. The terms “upstream” or “forward” refer to a direction against the flow direction.

(17) In addition, the term “adjacent to” may mean that an element is relatively near to but not in contact with a further element; or that the element is in contact with the further portion, unless the context clearly indicates otherwise. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. Terms “about” or “substantially” or like terms are intended to cover variations in a value that are within normal industry manufacturing tolerances

for that dimension. If no industry standard is available, a variation of twenty percent would fall within the meaning of these terms unless otherwise stated.

(18) FIG. 1 illustrates an example of a gas turbine engine **100** including a compressor section **102**, a combustion section **104**, and a turbine section **106** arranged along a central axis **112**. The compressor section **102** includes a plurality of compressor stages **114** with each compressor stage **114** including a set of stationary compressor vanes **116** or adjustable guide vanes and a set of rotating compressor blades **118**. A rotor **134** supports the rotating compressor blades **118** for rotation about the central axis **112** during operation. In some constructions, a single one-piece rotor **134** extends the length of the gas turbine engine **100** and is supported for rotation by a bearing at either end. In other constructions, the rotor **134** is assembled from several separate spools that are attached to one another or may include multiple disk sections that are attached via a bolt or plurality of bolts.

(19) The compressor section **102** is in fluid communication with an inlet section **108** to allow the gas turbine engine **100** to draw atmospheric air into the compressor section **102**. During operation of the gas turbine engine **100**, the compressor section **102** draws in atmospheric air and compresses that air for delivery to the combustion section **104**. The illustrated compressor section **102** is an example of one compressor section **102** with other arrangements and designs being possible.

(20) In the illustrated construction, the combustion section **104** includes a plurality of separate combustors **120** that each operate to mix a flow of fuel with the compressed air from the compressor section **102** and to combust that air-fuel mixture to produce a flow of high temperature, high pressure combustion gases or exhaust gas **122**. Of course, many other arrangements of the combustion section **104** are possible.

(21) The turbine section **106** includes a plurality of turbine stages **124** with each turbine stage **124** including a number of stationary turbine vanes **126** and a number of rotating turbine blades **128**. The turbine stages **124** are arranged to receive the exhaust gas **122** from the combustion section **104** at a turbine inlet **130** and expand that gas to convert thermal and pressure energy into rotating or mechanical work. The turbine section **106** is connected to the compressor section **102** to drive the compressor section **102**. For gas turbine engines **100** used for power generation or as prime movers, the turbine section **106** is also connected to a generator, pump, or other device to be driven. As with the compressor section **102**, other designs and arrangements of the turbine section **106** are possible.

(22) An exhaust portion **110** is positioned downstream of the turbine section **106** and is arranged to receive the expanded flow of exhaust gas **122** from the final turbine stage **124** in the turbine section **106**. The exhaust portion **110** is arranged to efficiently direct the exhaust gas **122** away from the turbine section **106** to assure efficient operation of the turbine section **106**. Many variations and design differences are possible in the exhaust portion **110**. As such, the illustrated exhaust portion **110** is but one example of those variations.

(23) A control system **132** is coupled to the gas turbine engine **100** and operates to monitor various operating parameters and to control various operations of the gas turbine engine **100**. In preferred constructions the control system **132** is typically micro-processor based and includes memory devices and data storage devices for collecting, analyzing, and storing data. In addition, the control system **132** provides output data to various devices including monitors, printers, indicators, and the like that allow users to interface with the control system **132** to provide inputs or adjustments. In the example of a power generation system, a user may input a power output set point and the control system **132** may adjust the various control inputs to achieve that power output in an efficient manner.

(24) The control system **132** can control various operating parameters including, but not limited to variable inlet guide vane positions, fuel flow rates and pressures, engine speed, valve positions, generator load, and generator excitation. Of course, other applications may have fewer or more controllable devices. The control system **132** also monitors various parameters to assure that the

gas turbine engine **100** is operating properly. Some parameters that are monitored may include inlet air temperature, compressor outlet temperature and pressure, combustor outlet temperature, fuel flow rate, generator power output, bearing temperature, and the like. Many of these measurements are displayed for the user and are logged for later review should such a review be necessary.

(25) FIG. 2 illustrates a perspective view of a turbine blade **200** that can be used in the gas turbine engine **100** in FIG. 1. A plurality of the turbine blades **200** may replace the rotating turbine blades **128** or combine with the rotating turbine blades **128**.

(26) The turbine blade **200** includes a platform **202**, a blade airfoil **204**, and a blade root **206**. The platform **202** has a first side **208** that faces away from the rotor **134** and a second side **210** that is opposite to the first side **208** and faces toward the rotor **134**.

(27) The blade airfoil **204** is attached to the platform **202** at the first side **208**. The blade airfoil **204** has a pressure side wall **212** and a suction side wall **214**. The pressure side wall **212** and the suction side wall **214** meet at a blade airfoil leading edge **216** and a blade airfoil trailing edge **218** with respect to a flow direction of a working flow **234**. The working flow **234** may be the exhaust gas **122** from the combustion section **104**. The pressure side wall **212** and the suction side wall **214** define a blade airfoil interior **220**.

(28) The blade root **206** is attached to the platform **202** at the second side **210**. The blade root **206** has a blade root leading edge **236** and a blade root trailing edge **238** with respect to the flow direction of the working flow **234**. The blade root **206** includes a plurality of cooling passages that are defined within the blade root **206**. In the construction illustrated in FIG. 2, the blade root **206** includes six cooling passages including, from the blade root leading edge **236** to the blade root trailing edge **238**, a first leading cooling passage **222**, a second leading cooling passage **224**, a first mid cooling passage **226**, a second mid cooling passage **228**, a first trailing cooling passage **230**, and a second trailing cooling passage **232**. The cooling passages extend to the platform **202** and are in flow communication with the blade airfoil interior **220** to provide cooling air to the plurality of cooling channels of the blade airfoil **204**. In other constructions, the blade root **206** may have more or less than six cooling passages.

(29) The turbine blade **200** includes a manifold **300** that is attached to the blade root **206** at a side facing to the rotor **134**. The manifold **300** is non-planar and meters or controls the quantity of cooling air that flows to each cooling passage of the plurality of cooling passages or blocks the cooling air from entering at least one of the cooling passages.

(30) FIG. 3 illustrates the manifold **300** for use with the turbine blade **200** of FIG. 2. The manifold **300** includes an outer wall **328** that can be made up of one or more features, walls, or plates, which define the non-planar manifold **300**. In the construction illustrated in FIG. 3, the outer wall **328** is made up of a first side plate **302**, a second side plate **304**, an outer plate **306**, a forward plate **308**, and an aft plate **310**. The outer plate **306** is solid and extends between the first side plate **302** and the second side plate **304**. The forward plate **308** is positioned at an upstream side with respect to the flow direction of the working flow **234**. The forward plate **308** is perpendicular to the outer plate **306**. The aft plate **310** is positioned at a downstream side with respect to the flow direction of the working flow **234**. The aft plate **310** is perpendicular to the outer plate **306**. The first side plate **302**, the second side plate **304**, the outer plate **306**, the forward plate **308**, and the aft plate **310** define a plenum therein. The forward plate **308**, the aft plate **310**, and the outer plate **306** are solid. A side that is opposite to the outer plate **306** is open and defines a perimeter having a planar surface that is attached to the blade root **206**. The forward plate **308** is disposed at the blade root leading edge **236** and the aft plate **310** is disposed at the blade root trailing edge **238** once the manifold **300** is attached to the blade root **206**.

(31) The manifold **300** includes a plurality of divider plates **312** that divide the manifold **300** into a plurality of compartments. Each divider plate **312** of the plurality of divider plates **312** is solid and perpendicular to the outer plate **306**. In the construction illustrated in FIG. 3, the manifold **300** includes five divider plates **312** that divide the manifold **300** into six compartments including, with

respect to the flow direction of the exhaust gas **122**, a first leading compartment **314**, a second leading compartment **316**, a first mid compartment **318**, a second mid compartment **320**, a first trailing compartment **322**, and a second trailing compartment **324**. In other constructions, the manifold **300** may have more or less than six compartments.

(32) The first side plate **302** and the second side plate **304** define a plurality of apertures **326**. In the construction shown in FIG. **3**, the first side plate **302** defines three apertures **326** that are open to the first leading compartment **314**. The second side plate **304** defines three apertures **326** that are open to the first leading compartment **314**. The six apertures **326** in the first leading compartment **314** equally contribute to a flow area of the first leading compartment **314**. The first side plate **302** defines one aperture **326** that is open to the second leading compartment **316**. The second side plate **304** defines one aperture **326** that is open to the second leading compartment **316**. The two apertures **326** in the second leading compartment **316** equally contribute a flow area of the second leading compartment **316**. The first side plate **302** defines one aperture **326** that is open to the first mid compartment **318**. The second side plate **304** defines one aperture **326** that is open to the first mid compartment **318**. The two apertures **326** in the first mid compartment **318** equally contribute a flow area of the first mid compartment **318**. The first side plate **302** defines six apertures **326** that are open to the second mid compartment **320**. The second side plate **304** defines six apertures **326** that are open to the second mid compartment **320**. The twelve apertures **326** in the second mid compartment **320** equally contribute a flow area of the second mid compartment **320**. The first side plate **302** and the second side plate **304** in the first trailing compartment **322** includes no apertures **326**. The first side plate **302** defines five apertures **326** that are open to the second trailing compartment **324**. The second side plate **304** defines five apertures **326** that are open to the second trailing compartment **324**. The ten apertures **326** in the second trailing compartment **324** equally contribute a flow area of the second trailing compartment **324**. The flow areas of the compartments may be different from each other. In general, the flow areas of the compartments at the blade root leading edge **236** is larger than the flow areas of the compartments at the blade root trailing edge **238**. Other constructions could include apertures **326** that extend through the outer wall **328** in any direction desired. For example, the apertures **326** may pass through the outer plate **306**.

(33) In the construction shown in FIG. **3**, the flow area of the second leading compartment **316** is larger than the flow area of the first leading compartment **314**, the flow area of the first mid compartment **318**, the flow area of the second mid compartment **320**, and the flow area of the second trailing compartment **324**. The flow area of the first mid compartment **318** is larger than the flow area of the first leading compartment **314**, the flow area of the second mid compartment **320**, and the flow area of the second mid second trailing compartment **324**. The flow area of the first leading compartment **314** is larger than the flow area of the second mid compartment **320** and the flow area of the second trailing compartment **324**. The flow area of the second mid compartment **320** is larger than the flow area of the second trailing compartment **324**.

(34) In other constructions, each compartment may have different numbers of apertures **326** than the manifold **300** shown in FIG. **3**. The flow area of each compartment may be different than the manifold **300** shown in FIG. **3**. The geometries of the apertures **326** may be different than the geometries of the apertures **326** shown in FIG. **3**.

(35) FIG. **4** illustrates a perspective exploded view of the manifold **300** shown in FIG. **3**. A plurality of gaps **402** are formed in the first side plate **302** and the second side plate **304**. The first side plate **302**, the second side plate **304**, and the outer plate **306** each have a generally rectangular shape. The first side plate **302** and the second side plate **304** perpendicularly extend from two sides of the outer plate **306**. Each of the first side plate **302**, the second side plate **304**, and the outer plate **306** has rounded edges. The first side plate **302**, the second side plate **304**, and the outer plate **306** are formed as a single piece. In other constructions, the first side plate **302**, the second side plate **304**, and the outer plate **306** may be formed as separated pieces and assembled together.

(36) The forward plate **308** has a general square shape having rounded edges. The forward plate

308 is perpendicularly welded or brazed to the first side plate **302**, second side plate **304**, and the outer plate **306**.

(37) The aft plate **310** has a general square shape having rounded edges. The aft plate **310** is perpendicularly welded or brazed to the first side plate **302**, second side plate **304**, and the outer plate **306**. The aft plate **310** is thicker than the forward plate **308**. In other constructions, the aft plate **310** may be thinner or equal to the forward plate **308**.

(38) Each divider plate **312** has a general square shape having rounded edges. Each divider plates **312** has a cutout **404** to engage with the outer plate **306**. Each divider plate **312** is inserted into one of the gaps **402** and welded or brazed to the first side plate **302**, the second side plate **304**, and the outer plate **306**.

(39) FIG. 5 illustrates a perspective view of a manifold **500** suitable for use with the turbine blade **200** shown in FIG. 2. The manifold **500** may replace the manifold **300** shown in FIG. 2.

(40) In the construction shown in FIG. 5, the forward plate **308** and the divider plate **312** are oblique to the outer plate **306**. It is also possible that the aft plate **310** is oblique to the outer plate **306**. It is also possible that is oblique to the outer plate **306**. The manifold **500** may be manufactured as a single piece using methods, such as additive manufacture. The manifold **500** otherwise has the similar configuration as the manifold **300**.

(41) FIG. 6 illustrates a perspective view of a manifold **600** suitable for use with the turbine blade **200** shown in FIG. 2. The manifold **600** may replace the manifold **300** shown in FIG. 2.

(42) In the construction shown in FIG. 6, the apertures **326** are formed at the outer plate **306** and the forward plate **308**. It is also possible that the apertures **326** are formed at the aft plate **310**. The manifold **500** otherwise has the similar configuration as the manifold **300**.

(43) FIG. 7 illustrates a portion of a perspective view of the turbine blade **200** shown in FIG. 2 that better illustrates the blade root **206** and the manifold **300**. The manifold **300** is aligned with the blade root **206** with a position pin **702** that is inserted into the blade root **206**. The position pin **702** is inserted into the blade root **206** through the first trailing compartment **322**. The manifold **300** is perimeter welded to a blade root end surface **706** all around, using, for example, a fillet weld. The outer plate **306** is spaced a non-zero distance away from the blade root end surface **706**.

(44) IG. 8 illustrates a perspective view of a manifold **800** suitable for use with the turbine blade **200** shown in FIG. 2.

(45) In the construction shown in FIG. 8, the manifold **800** includes a base plate **802**. The base plate **802** is one single piece. The first leading compartment **314**, the second leading compartment **316**, the first mid compartment **318**, the second mid compartment **320**, the first trailing compartment **322**, and the second trailing compartment **324** are formed at the base plate **802** and extend out from the base plate **802**.

(46) Each of the compartments has a compartment outer wall **804**. The compartment outer wall **804** may have a compartment outer plate **806** that is spaced a non-zero distance away from the base plate **802**. Apertures **326** are defined at the compartment outer wall **804**. As illustrated in FIG. 8, the second mid compartment **320** has at least one aperture **326** that is formed at the compartment outer wall **804** of the second mid compartment **320**. The first trailing compartment **322** has at least one aperture **326** that is formed at the compartment outer wall **804** of the first trailing compartment **322**. The second trailing compartment **324** has at least one aperture **326** that is formed at the compartment outer wall **804** of the second trailing compartment **324**. The apertures **326** are formed at the side plates of the compartment outer wall **804**. The compartment outer walls **804** of the first leading compartment **314**, the second leading compartment **316**, and the first mid compartment **318** are solid.

(47) In other constructions, the first leading compartment **314**, the second leading compartment **316**, and the first mid compartment **318** may have at least one aperture **326** that is formed at the compartment outer wall **804** of the respective first leading compartment **314**, the second leading compartment **316**, and the first mid compartment **318**. It is also possible that the second mid

compartment **320**, the first trailing compartment **322**, and the second trailing compartment **324** are solid. It is also possible that the apertures **326** are formed at the compartment outer plate **806**.

(48) FIG. **9** illustrates a perspective view of a manifold **900** suitable for use with the turbine blade **200** shown in FIG. **2**.

(49) The manifold **900** has a plurality of base plates. The plurality of base plates are separated pieces from each other. Each of the first leading compartment **314**, the second leading compartment **316**, the first mid compartment **318**, the second mid compartment **320**, the first trailing compartment **322**, and the second trailing compartment **324** is formed at one of the plurality of base plates and extends out from the respective base plate.

(50) For example, in the construction shown in FIG. **9**, the manifold **900** has a first base plate **902**, a second base plate **904**, and a third base plate **906**. The first base plate **902**, the second base plate **904**, and the third base plate **906** are separated pieces. The second base plate **904** is disposed between the first base plate **902** and the third base plate **906**. The second base plate **904** has an arm **908** that extends out from the second base plate **904** and engages with a recess **910** of the third base plate **906**. In other constructions, the third base plate **906** may have an arm that extends out from the third base plate **906** and engages with a recess of the second base plate **904**. It is also possible that the first base plate **902** has an arm that extends out from the first base plate **902** and engages with a recess of the second base plate **904**, or vice versa.

(51) The first leading compartment **314** is formed at the first base plate **902** and extends out from the first base plate **902**. The second leading compartment **316** is formed at the second base plate **904** and extends out from the second base plate **904**. The first mid compartment **318** is formed at the third base plate **906** and extends out from the third base plate **906**. The first leading compartment **314** has at least one aperture **326** that is formed at the compartment outer wall **804** of the first leading compartment **314**. The second leading compartment **316** has at least one aperture **326** that is formed at the compartment outer wall **804** of the second leading compartment **316**. The first mid compartment **318** has at least one aperture **326** that is formed at the compartment outer wall **804** of the first mid compartment **318**.

(52) Further compartments, such as the second mid compartment **320**, the first trailing compartment **322**, and the second trailing compartment **324** may be each formed at a further base plate. It is possible that at least one of the compartment outer walls **804** is solid.

(53) FIG. **10** illustrates a portion of a perspective view of the turbine blade **200** shown in FIG. **2** that better illustrates the blade root **206** and the manifold **800**.

(54) In the construction shown in FIG. **10**, the base plate **802** is attached to the blade root **206**. The first leading compartment **314** is inserted into the first leading cooling passage **222**. The second leading compartment **316** is inserted into the second leading cooling passage **224**. The first mid compartment **318** is inserted into the first mid cooling passage **226**. The second mid compartment **320** is inserted into the second mid cooling passage **228**. The first trailing compartment **322** is inserted into the first trailing cooling passage **230**. The second trailing compartment **324** is inserted into the second trailing cooling passage **232**. The compartment outer plate **806** is spaced a non-zero distance away from the blade root end surface **706**.

(55) The manifold **800** may be replaced by the manifold **900** to be attached to the blade root **206**. Each separated base plate is attached to the blade root **206** and the compartment that is formed at the base plate is inserted into one of the cooling passages. For example, the first base plate **902** is attached to the blade root **206** and the first leading compartment **314** that is formed at the first base plate **902** is inserted into the first leading cooling passage **222**. The second base plate **904** is attached to the blade root **206** and the second leading compartment **316** that is formed at the first base plate **902** is inserted into the second leading cooling passage **224**. The third base plate **906** is attached to the blade root **206** and the first mid compartment **318** that is formed at the third base plate **906** is inserted into the first mid cooling passage **226**. Further base plates may be attached to the blade root **206** and further compartments each formed at one of the further base plates may be

inserted into one of the second mid cooling passage **228**, the first trailing cooling passage **230**, and the second trailing cooling passage **232**.

(56) In other constructions, the manifold **800** and manifold **900** may be attached to the blade root **206** in a way such that the compartments extend out a non-zero distance away from the cooling passages, respectively.

(57) In operation, with reference to FIG. **2** through FIG. **10**, the manifolds **300**, the manifold **500**, the manifold **600**, the manifold **800**, or the manifold **900** is attached to the blade root **206** with the forward plate **308** disposed at the blade root leading edge **236** and the aft plate **310** disposed at the blade root trailing edge **238**. Cooling air **704** enters the manifold and is throttled through the apertures **326** while entering. By placing the outer plate **306** or the compartment outer plate **806** away from the blade root end surface **706** with a non-zero distance, the cooling air **704** enters and is mixed in the first leading compartment **314**, the second leading compartment **316**, the first mid compartment **318**, the second mid compartment **320**, and the second trailing compartment **324** before entering the first leading cooling passage **222**, the second leading cooling passage **224**, the first mid cooling passage **226**, the second mid cooling passage **228**, and the second trailing cooling passage **232**, respectively. The compartments create a plenum for the cooling air **704** before flowing into the cooling passages. The manifold is thus functioned as a flow conditioner such that the inlets of the cooling passages are shielded from throttling disturbance that may be created by the various apertures **326**. The uniform and diffused cooling air **704** reduces asymmetric heat transfer.

(58) The cooling air **704** enters the manifold **300**, or the manifold **500** from the first side plate **302** and the second side plate **304** which generate effective mixing before entering the cooling passages. The cooling air **704** turns 90 degrees in the compartments and flows into the cooling passages to reduce the cooling air **704** bias from the downstream side to the upstream side.

(59) The first leading compartment **314** meters a non-zero first flow of cooling air **704** to the first leading cooling passage **222** through the flow area of the apertures **326** in the first leading compartment **314**. The second leading compartment **316** meters a non-zero second flow of cooling air **704** to the second leading cooling passage **224** through the flow area of the apertures **326** in the second leading compartment **316**. The first mid compartment **318** meters a non-zero third flow of cooling air **704** to the first mid cooling passage **226** through the flow area of the apertures **326** in the first mid compartment **318**. The second mid compartment **320** meters a non-zero fourth flow of cooling air **704** to the second mid cooling passage **228** through the flow area of the apertures **326** in the second mid compartment **320**. The first trailing compartment **322** has no apertures **326** so that the cooling air **704** is blocked from entering the first trailing cooling passage **230**. The second trailing compartment **324** meters a non-zero fifth flow of cooling air **704** to the second trailing cooling passage **232** through the apertures **326** in the second trailing compartment **324**. The non-zero first flow of cooling air **704**, the non-zero second flow of cooling air **704**, the non-zero third flow of cooling air **704**, the non-zero fourth flow of cooling air **704**, and the non-zero fifth flow of cooling air **704r** may be different from each other.

(60) Discrete throttling control (i.e., by selecting aperture flow areas) of the flow of cooling air **704** that flows to each cooling passage is achieved by different arrangements of the compartments. In general, the flow areas of the apertures **326** in the compartments at the blade root leading edge **236** is larger than the flow areas of the apertures **326** in the compartments at the blade root trailing edge **238**. For example, in manifold **300** and manifold **500**, the flow area of the apertures **326** in the second leading compartment **316** and the flow area of the apertures **326** in the first mid compartment **318** is larger than the flow area of the apertures **326** in the first leading compartment **314**, the flow area of the apertures **326** in the second mid compartment **320**, and the flow area of the apertures **326** in the second trailing compartment **324**. As such, the cooling air **704** is the least throttled in the second leading compartment **316** and the first mid compartment **318** than in the first leading compartment **314**, the second mid compartment **320**, and the second trailing compartment

324. The first trailing compartment 322 has no apertures 326 such that the cooling air 704 is blocked from entering the first trailing cooling passage 230.

(61) The manifolds better controls the cooling air 704 flowing into the cooling passages which allows more accurate prediction for the life of the blade. In addition, the manifolds assist in reducing leakages of the cooling air 704.

(62) Although an exemplary embodiment of the present disclosure has been described in detail, those skilled in the art will understand that various changes, substitutions, variations, and improvements disclosed herein may be made without departing from the spirit and scope of the disclosure in its broadest form.

(63) None of the description in the present application should be read as implying that any particular element, step, act, or function is an essential element, which must be included in the claim scope: the scope of patented subject matter is defined only by the allowed claims. Moreover, none of these claims are intended to invoke a means plus function claim construction unless the exact words “means for” are followed by a participle.

Claims

1. A turbine blade comprising: a platform defining a first side and a second side; a blade airfoil attached to the first side of the platform, the blade airfoil comprising a pressure side wall and a suction side wall defining a blade airfoil interior; a blade root attached to the second side of the platform, the blade root including a blade root leading edge and a blade root trailing edge with respect to a flow direction of a working flow, the blade root defining a first cooling passage and a second cooling passage that are in flow communication with the blade airfoil interior; and a manifold attached to the blade root, the manifold being non-planar, the manifold comprising: an outer wall; a first compartment having a first flow area defined by a first aperture, the first aperture being formed at the outer wall and open to the first compartment, the first compartment metering a non-zero first flow of cooling air to the first cooling passage through the first flow area; and a second compartment having a second flow area defined by a second aperture, the second aperture being formed at the outer wall and open to the second compartment, the second compartment metering a non-zero second flow of cooling air to the second cooling passage through the second flow area that is different than the first flow area, wherein the outer wall further comprises: a first side plate that extends between the blade root leading edge and the blade root trailing edge; a second side plate that extends between the blade root leading edge and the blade root trailing edge; an outer plate that extends between the first side plate and the second side plate; a forward plate attached to the first side plate, the second side plate, and the outer plate at one end; and an aft plate attached to the first side plate, the second side plate, and the outer plate at the other end that is opposite to the one end, wherein the outer plate is solid and spaced a non-zero distance away from a blade root end surface.
2. The turbine blade of claim 1, wherein the manifold comprises a divider plate that is disposed between the forward plate and the aft plate, and wherein the divider plate divides the manifold into the first compartment and the second compartment.
3. The turbine blade of claim 1, wherein the first aperture is formed at the first side plate.
4. The turbine blade of claim 1, wherein the first compartment comprises a plurality of first apertures that are formed at the outer wall, and wherein the plurality of first apertures define the first flow area.
5. The turbine blade of claim 1, wherein the blade root defines a third cooling passage that is in flow communication with the blade airfoil interior, wherein the manifold comprises a third compartment, wherein a portion of the outer wall that defines the third compartment is solid, and wherein the third compartment blocks the cooling air from entering the third cooling passage.

6. The turbine blade of claim 1, wherein a perimeter of the manifold defines a planar surface arranged to attach to the blade root.
